# Tutorial 03

## Measurable Map and Discrete Probability Measures

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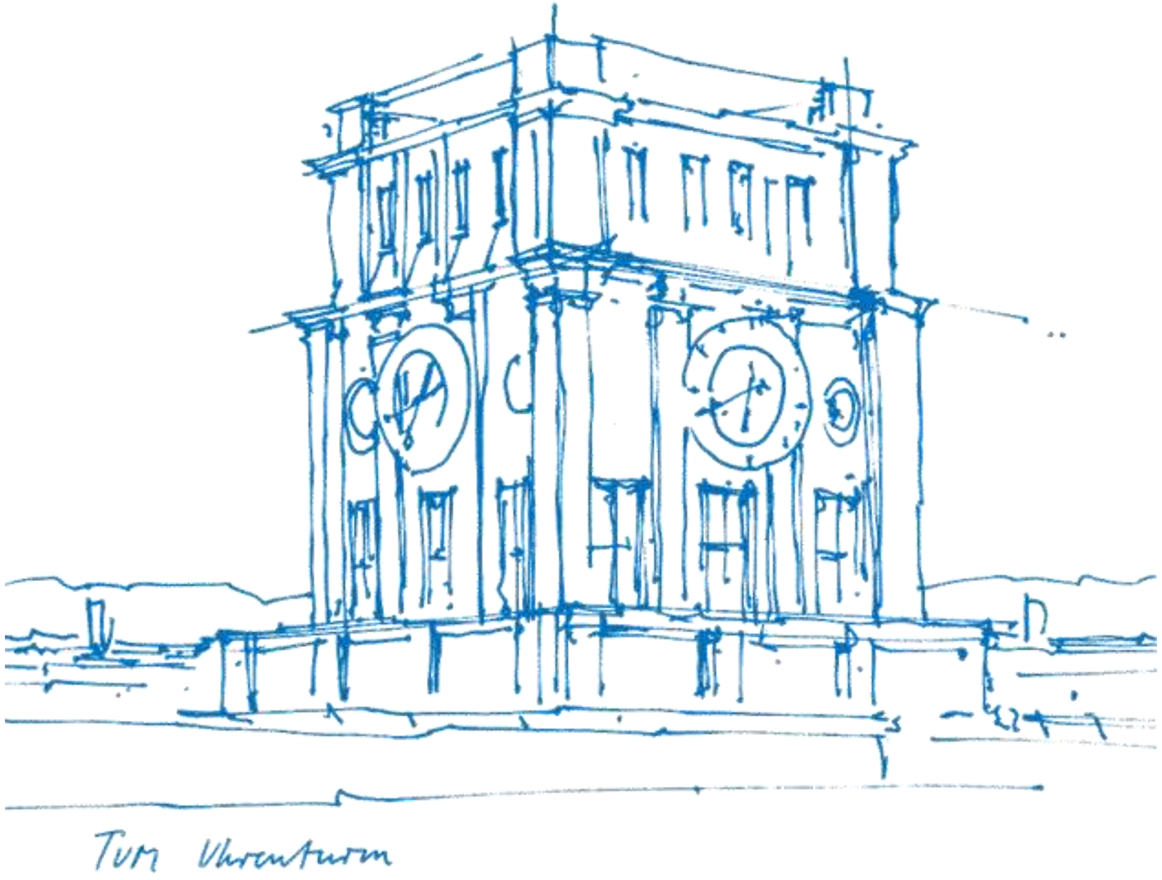

# About Tutorial

- **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
  - 14:15-15:45 Tue, 00.08.053 (B42)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code ☞

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➢ PART I. Recap of the Lecture

➢ PART II. Exercises from the Exercise Sheet

➢ PART III. Additional Exercises (with solution)

# I. Recap

- **σ-Algebra (Review)**
  - Generator of σ-algebra
  - Induced σ-algebra

- **Measurable Map and Random Variable**
  - Measurable map
  - (Real-valued) random variable

- **Discrete Probability Measures**
  - Common measures
    - Bernoulli process
    - Binomial distribution
    - Geometric distribution
    - Poisson distribution

# II. Exercise Sheet

**Exercise 1.** Let $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$ be measurable spaces, $f : X \to Y$ a map, and $\mathcal{E}_Y \subseteq \mathcal{A}_Y$ arbitrary. Show that

$$\sigma(f^{-1}(\mathcal{E}_Y)) = f^{-1}(\sigma(\mathcal{E}_Y)) .$$

Further, show that

$$f \text{ is } \mathcal{A}_X\text{-}\sigma(\mathcal{E}_Y)\text{-measurable} \quad \Longleftrightarrow \quad f^{-1}(E) \in \mathcal{A}_X \text{ for all } E \in \mathcal{E}_Y .$$

With that, finally show that for $\sigma(\mathcal{E}_Y) = \mathcal{A}_Y$ it holds that

$$f \text{ is } \mathcal{A}_X\text{-}\mathcal{A}_Y\text{-measurable} \quad \Longleftrightarrow \quad f^{-1}(\mathcal{E}_Y) \subseteq \mathcal{A}_X .$$

This means that to show that a function is measurable, it is enough to show measurability on a generator of the codomain $\sigma$-algebra.

# II. Exercise Sheet

**Exercise 2.** Let $\Omega \neq \emptyset$ be a set, $A \subseteq \Omega$, $A \neq \emptyset$, and let $\mathcal{E} \subseteq \mathcal{P}(\Omega)$. Then it holds that

$$\sigma(\mathcal{E}\big|_A) = \sigma(\mathcal{E})\big|_A .$$

# II. Exercise Sheet

**Exercise 3.** Let $(p_n)_{n\in\mathbb{N}} \subset [0,1]$ with a fixed $\lambda := \lim_{n\to\infty} n\, p_n \in \mathbb{R}_{>0}$. Then for all $\omega \in \mathbb{N}$ we have that

$$\lim_{n\to\infty} p_{\mathrm{Bin}(n,p_n)}(\omega) = p_{\mathrm{Poi}(\lambda)}(\omega)\,.$$

# II. Exercise Sheet

**Exercise 4.** After a long game night, Agathe and Balthasar are standing alone at the subway station Marienplatz, where a train departs every 10 minutes. Unfortunately, the trains are so overcrowded that only one person can board each time. Every person on the platform has the same chance of boarding the train. Additionally, until the next train arrives, two new people arrive at the station. Determine the probability that Agathe will be able to board

(i) the $n$-th train,

(ii) at the latest by the $n$-th train,

where $n \in \mathbb{N}$.

# III. Additional Exersices

**Exercise 5.** Let $X$ be a Poisson-distributed random variable with parameter $\lambda \geq 0$. Which equation does the probability mass function $f_X$ satisfy for all $i \in \mathbb{N}_0$?

✓ – $f_X(i+1) = \frac{\lambda}{i+1} \cdot f_X(i)$

– $f_X(i+1) = \frac{\lambda}{i} \cdot f_X(i)$

– $f_X(i+1) = \frac{e^{-\lambda}\lambda}{i+1} \cdot f_X(i)$

– $f_X(i+1) = \frac{e^{-\lambda}}{i+1} \cdot f_X(i)$

$$f_X(i) = \frac{e^{-\lambda}\lambda^i}{i!}, \quad f_X(i+1) = \frac{e^{-\lambda}\lambda^{i+1}}{(i+1)!} = \frac{\lambda}{(i+1)} \cdot \frac{e^{-\lambda}\lambda^i}{i!} = \frac{\lambda}{(i+1)} f_X(i).$$

# Materials